Safa Paravakkal

Biomedical Sciences Student · Neuroscience, Digital Health & Research Software

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PROFILE

Second-year Biomedical Sciences student at Royal Holloway with a strong interest in neuroscience, digital health, and research software. I'm building tools for searching, analysing, and appraising biomedical research, including a medical evidence platform and a clinical trials dashboard using the ClinicalTrials.gov API. Through these projects, I've developed practical experience in Python, APIs, data workflows, testing, and scientific literature analysis. I'm seeking internships or placement opportunities where I can contribute to health tech, research, or data-driven product teams.

PROJECTS

NeuroTrials Explorer – Data Dashboard · Solo Project *April 2026 · Completed*

An interactive Streamlit dashboard exploring clinical trial trends across 7 neurological conditions using the ClinicalTrials.gov API · Python · Requests · Pandas · Streamlit · Plotly

- Designed and iterated on an ETL workflow for clinical trial records across 7 neurological conditions
- Used Python and pandas to clean and reshape nested trial data,
- Implemented and customised 6 interactive Plotly Express charts to explore trial patterns by condition, phase, sponsor, status, and year
- Deployed on Streamlit Community Cloud

Medical Evidence Platform (Seren) – Solo Project *March 2026 · Live and under active development*

A full-stack platform for discovering, importing, and AI-analysing peer-reviewed medical research · Python · FastAPI · SQL Model · SQLite · pytest · GitHub Actions

- Built a unified search workflow across Europe PMC, OpenAlex, Crossref, and Semantic Scholar
- Developed provider parsing and deduplication logic using DOI matching, metadata comparison, and cross-source reference tracking
- Implemented search filters for study type, including RCTs, meta-analyses, cohort studies, and systematic reviews
- Added AI-powered research analysis features with model routing and fallback logic
- Maintained a 93-test suite covering provider parsing, authentication, deduplication, and study workflows
- Set up CI/CD with GitHub Actions to run tests automatically on each push

EXPERIENCE

Sales Advisor – H&M — Sloane Square & Kensington *August 2024 – June 2025*

- Customer service in a high-footfall flagship store, maintaining standards during regional management visits
- Supported stock control, replenishment, and visual merchandising
- Built strong communication, reliability, and time-management skills through continuous customer-facing work

Volunteer – Nettlestead Care Home *May 2023*

- Supported residents and staff, gaining exposure to patient-facing communication and care settings

EDUCATION

BSc Biomedical Sciences – Royal Holloway, University of London *2024 – Present · Second Year*

A Levels – Bullers Wood School for Girls *2022 – 2024*

Chemistry · Biology · Psychology

GCSEs · 9 Subjects – Coopers Secondary School *2017 – 2022*

Incl. English, Maths, Triple Science, History, Art

SKILLS

Technical: Python · FastAPI · pandas · Streamlit · Plotly · REST APIs · SQLModel · SQLite · pytest · GitHub Action · JavaScript · HTML/CSS

Scientific & Domain: Clinical evidence hierarchy · RCT & meta-analysis appraisal · PICO framework · Biomedical research methods · Medical literature databases

Professional: Independent project delivery · Problem-solving & research · Team collaboration· Time management

ADDITIONAL

Certifications: Google Data Analytics — Coursera (in progress)

Availability: Immediately available — internships, work experience, short-term placements